

Problem

Design a problem, to be solved using *PSpice* or *MultiSim*, to help other students better understand source-free *RLC* circuits.

Step-by-step solution

Step 1 of 1

Design a problem, using PSpice, to help other students better understand source-free *RLC* circuits.

Although there are many ways to work this problem, this is an example based on a somewhat similar problem worked in the third edition.

Problem

The step response of an *RLC* circuit is given by

$$\frac{d^2 i_L}{dt^2} + 0.5 \frac{di_L}{dt} + 4i_L = 0$$

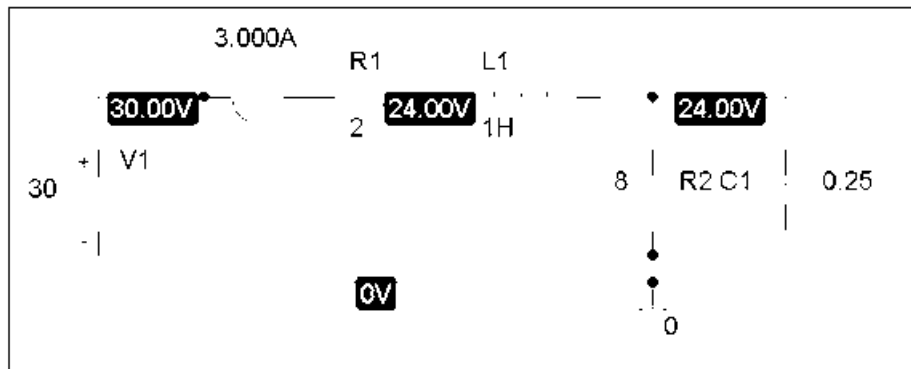
Given that $i_L(0) = 3$ A and $v_C(0) = 24$ V, solve for $v_C(t)$ and $i_C(t)$.

Solution

(a) For $t > 0$, we have the schematic below. When this is saved and simulated, we obtain the initial inductor current and capacitor voltage as

$i_L(0) = 3$ A and $v_C(0) = 24$ V.

$$i_L(0) = 3 \text{ A and } v_C(0) = 24 \text{ V.}$$



(b) For $t < 0$, we have the schematic shown below. To display $i(t)$ and $v(t)$, we insert current and voltage markers as shown. The initial inductor current and capacitor voltage are also incorporated. In the Transient box, we set Print Step = 25 ms and the Final Time to 4s. After simulation, we automatically have $i_o(t)$ and $v_o(t)$ displayed as shown below.

